

Implicit assessment in psychology: How to work with the IAT

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What we're using today

- download  and follow the instructions
- download  and follow the instructions
-  for an easy analysis of the data (no need to install)
- `implicitMeasures` R package

Disclaimer

This tutorial assumes that Inquisit is used for collecting the IAT data.

It is indeed based on data collected with Inquisit, available [here](#)

It is a Race IAT data set with observations from 142 participants.

Explicit measures (i.e., political orientation and attitudes towards Black/White people) have been collected as well

Both the shiny app and the package can be used with data collected with other administration software

- 1 DScoreApp
- 2 `implicitMeasures`
- 3 Supplementary material

If you want it simple, DScoreApp is made for you!

Advantages

- It's super easy to use
- Well documented and the maintainer is always available to help you
- You can visually inspect the results as you compute them
- There's a toy data set with which you can familiarize with the app

Disadvantages

- You have to use excel to prepare the data
- You can compute only one D score at the time
- You have to manually merge in excel the data set with explicit measures to run further analysis

Let's start

Let's take a look at [DScoreApp](#)

Let's start

Let's take a look at [DScoreApp](#)

Data set have to be arranged in 4 columns, named:

- **participant**: participant's IDs
- **block**: Mapping A and Mapping B block labels
- **latency**: response times
- **correct**: accuracy responses

Date	Time	build	subject	blocknum	blockcode	trialnum	trialcode	response	correct	latency	stimulusitem
12/31/18	09:55	3.0.e.o.1	1		consenso	1	consenso	Disponibile a partecipare alla ricerca nei termini sopra indicati	1	27280	0
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	1	reminder	57	401	Controlla le categorie - premi SPASIO per cominciare	
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	2	PlainWhiteLeft	18	1	524	wf6.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	4	PlainBlackRight	23	1	637	bmf6.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	6	PlainWhiteLeft	18	1	730	wm.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	8	PlainBlackRight	23	1	550	bf14.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	10	PlainBlackRight	23	1	588	bm14.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	12	PlainBlackRight	23	1	736	bm23.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	14	PlainWhiteLeft	18	1	612	wf3.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	16	PlainWhiteLeft	23	0	337	wm6.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	18	PlainWhiteLeft	18	1	1390	wm.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	20	PlainWhiteLeft	18	1	1255	wm4.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	22	PlainBlackRight	23	1	704	bf96.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	24	WhiteBlack	23	0	536	bf23.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	26	PlainBlackRight	23	1	762	bf14.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	28	PlainBlackRight	23	1	642	bm14.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	30	PlainBlackRight	23	1	1056	bf56.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	32	PlainBlackRight	23	1	2043	bm56.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	34	PlainWhiteLeft	18	1	967	wm6.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	36	PlainWhiteLeft	18	1	824	wf2.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	38	PlainWhiteLeft	18	1	1139	wm1.jpg
12/31/18	09:55	3.0.e.o.1	15		WhiteBlack	40	PlainWhiteLeft	18	1	719	wf3.jpg
12/31/18	09:55	3.0.e.o.1	21		badgood 1	reminder	57	1	3234	Controlla le categorie - premi SPASIO per cominciare	
12/31/18	09:55	3.0.e.o.1	21		badgood 2	PlainBlackLeft	18	1	1090	cattivo	
12/31/18	09:55	3.0.e.o.1	21		badgood 4	PlainBlackLeft	18	1	675	fallimento	

8 / 51

Just delete the unnecessary columns:

- `date`
- `time`
- `build`

The first column of interest is `blockcode`. Use the `filter` function to see all the possible values of the column

blockcode

The screenshot shows the 'blockcode' filter menu in the DScoreApp. The menu is open, displaying various options for filtering and sorting data. The background shows a data table with columns A through L.

Menu Options:

- Ordina dalla A alla Z
- Ordina dalla Z alla A
- Ordina per colore
- Visualizzazione foglio
- Cancello filtro da "blockcode"
- Filtra per colore
- Filtri per testo
- Cerca
- ☒ (Seleziona tutto)
- ☒ badgood
- ☒ consenso
- ☒ demografica
- ☒ PracticeWhitebad
- ☒ PracticeWhitegood
- ☒ TestWhitebad
- ☒ TestWhitegood
- ☒ WhiteBlack

Data Table (Columns A-L):

A	B	C	D	E	F	G	H	I	J	K	L
subject	blockn	blockcode	trialnu	trialcod	responsi	correct	latency	stimul	em1		
1	Qrdina dalla A alla Z		1	consenso	Disponibil	1	27280	0			
2	Qrdina dalla Z alla A		1	reminder	57	1	4016	Controlla le categorie - premi SPAZIO			
3	Ordina per colore		2	PlainWhit	18	1	924	wf6.jpg			
4	Visualizzazione foglio		4	PlainBlack	23	1	637	bm56.jpg			
5	Cancello filtro da "blockcode"		6	PlainWhit	18	1	732	wm1.jpg			
6	Filtra per colore		8	PlainBlack	23	1	550	bf14.jpg			
7	Filtri per testo		10	PlainBlack	23	1	588	bm14.jpg			
8	Cerca		12	PlainBlack	23	1	736	bm23.jpg			
9	☒ (Seleziona tutto)		14	PlainWhit	18	1	612	wf3.jpg			
10	☒ badgood		16	PlainWhit	23	0	837	wm6.jpg			
11	☒ consenso		18	PlainWhit	18	1	1380	wf2.jpg			
12	☒ demografica		20	PlainWhit	18	1	1255	wm4.jpg			
13	☒ PracticeWhitebad		22	PlainBlack	23	1	704	bf56.jpg			
14	☒ PracticeWhitegood		24	PlainBlack	23	1	636	bf23.jpg			
15	☒ TestWhitebad		26	PlainBlack	23	1	762	bf14.jpg			
16	☒ TestWhitegood		28	PlainBlack	23	1	641	bm14.jpg			
17	☒ WhiteBlack		30	PlainBlack	23	1	1096	bf56.jpg			
18	OK		32	PlainBlack	23	1	1033	bm56.jpg			
19	Annulla		34	PlainWhit	18	1	967	wm4.jpg			
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File Information:

- File: psiColalAtdata.csv
- Size: 548.3 KB
- Date: May 17, 2022, 10:01 AM

blockcode

The blocks we need are the “critical” blocks of the IAT:

- PracticeWhitegood
- TestWhitegood
- PracticeWhitebad
- TestWhitebad

blockcode

The blocks we need are the “critical” blocks of the IAT:

- PracticeWhitegood
- TestWhitegood
- PracticeWhitebad
- TestWhitebad

White-Good/Black-Bad Condition
(MappingA)

blockcode

The blocks we need are the “critical” blocks of the IAT:

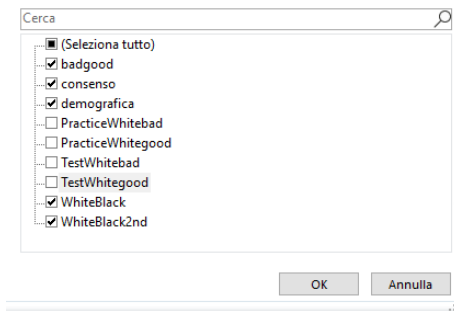
- PracticeWhitegood
- TestWhitegood
- PracticeWhitebad
- TestWhitebad

White-Good/Black-Bad Condition
(MappingA)

Black-Good/White-Bad Condition
(MappingB)

Just select those. . . . by eliminating all other blocks!

From the filter on the column blockcode, select all blocks but PracticeWhitegood, TestWhitegood, PracticeWhitebad, TestWhitebad:



blockcode

Highlight and delete all the rows that remained after the filter has been applied

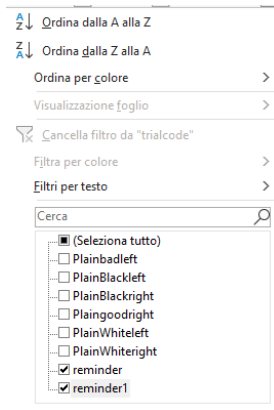
Don't panic when you see you have no more rows

Remove the filter from blockcode and it's all done...ish

trialcode

After selecting the blocks of interest, we still have some trials that are part of the experiment but of no interest for the D score computation (e.g., reminders, instructions)

Set a filter on the `trialcode` column and select the non-relevant trials:



trialcode

Select all the rows left after applying the filter

Delete them

(Again, don't panic)

Remove the filter from the `trialcode` column

Polish the data set

Remove any filter left

Remove all the unnecessary columns (`blocknum`, `trialnum`, `trialcode`, `response`, `stimulusitem1`)

Rename the remaining columns according to the shiny app instructions:

- `subject` → `participant`
- `blockcode` → `block`
- `latency` → `latency`
- `correct` → `correct`

The final look

	A	B	C	D	E
1	participant	block	correct	latency	
2	1	PracticeWhitebad	1	725	
3	1	PracticeWhitebad	1	1052	
4	1	PracticeWhitebad	1	1517	
5	1	PracticeWhitebad	1	767	
6	1	PracticeWhitebad	1	985	
7	1	PracticeWhitebad	1	708	
8	1	PracticeWhitebad	1	689	
9	1	PracticeWhitebad	1	719	
10	1	PracticeWhitebad	1	550	
11	1	PracticeWhitebad	1	1101	
12	1	PracticeWhitebad	1	918	
13	1	PracticeWhitebad	1	812	
14	1	PracticeWhitebad	1	717	
15	1	PracticeWhitebad	1	1028	
16	1	PracticeWhitebad	1	823	
17	1	PracticeWhitebad	1	843	
18	1	PracticeWhitebad	1	764	
19	1	PracticeWhitebad	1	651	
20	1	PracticeWhitebad	1	1076	

The file must be saved as `.csv` with “,” specified as the column separator

It's super important that “,” is set as the column separator because otherwise the app won't work and won't throw an error (because the maintainer thought “I'll add the error message tomorrow”)

Upload the data set

The screenshot shows the DScoreApp interface. On the left, there's a 'Choose CSV file' section with a 'Browse...' button and a 'No file selected' message. Below this are two mapping sections: 'MappingA' and 'MappingB', each with a 'Practice block label' and a 'Test block label' dropdown menu. At the bottom of this section are 'Persist Data' and 'Show info' buttons, and a 'WAITING FOR DATA' status message. On the right, there's a navigation menu with links: 'Read Me First', 'D-Score results', and 'Descriptive Statistics'. Below these are links for 'THE D-SCORE SHINY APP', 'IMPORT DATA', 'HOW IT WORKS', 'THE D-SCORE RESULTS PANEL', 'DESCRIPTIVE STATISTICS PANEL', 'WHAT YOU GET', 'REFERENCES', 'CONTACTS', and 'LICENSE'.

Browse through your files, select the file you have just created. It will be automatically uploaded

Prepare the data set

Example data

☐ Race IAT dataset

Choose CSV file

Discover... dataset.csv

MappingA Practice block label
e.g. practiceWhiteGood

PracticeWhitebad

MappingA Test block label
e.g. testWhiteGood

TestWhitebad

MappingB Practice block label
e.g. practiceWhiteBad

PracticeWhitegood

MappingB Test block label
e.g. testWhiteBad

TestWhitegood

PREPARE DATA

Show info

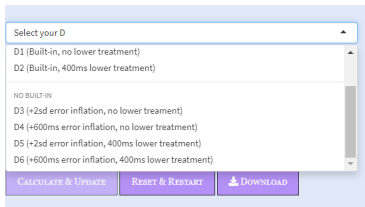
WAITING FOR DATA

To change the mapping for the computation of the D score, just select the labels of the blocks from the drop down menu

When you're all set just click "Prepare data" and wait for the "Data are ready" message to appear.

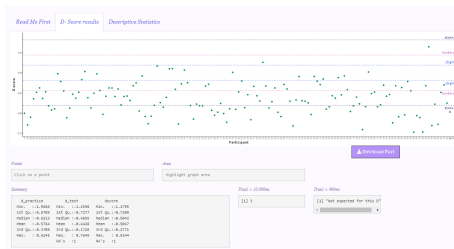
Select your D

Just select the D score you want to compute from the drop down menu, click on “Calculate & Update”, and it’s all done! The D scores will shortly appear in the Results panel



The screenshot shows a web application interface with a light blue background. At the top, there is a dropdown menu labeled "Select your D" with a small upward arrow on the right. The menu is open, displaying a list of options: "D1 (Built-in, no lower treatment)", "D2 (Built-in, 400ms lower treatment)", a separator line, "NO BUILT-IN", "D3 (+2sd error inflation, no lower treatment)", "D4 (+600ms error inflation, no lower treatment)", "D5 (+2sd error inflation, 400ms lower treatment)", and "D6 (+600ms error inflation, 400ms lower treatment)". Below the dropdown menu, there are three purple buttons with white text: "CALCULATE & UPDATE", "RESET & RESTART", and "DOWNLOAD" (which includes a download icon).

(default) Results panel



Play with graphical representations and settings

Accuracy deletion

☒ No
☐ Yes (Practice + Test blocks)

Fast participants deletion

☒ No
☐ Yes

Graphic display

☒ Points
☐ Histogram
☐ Density
☐ Histogram + Density

Point Graph

None ▼

Note: Please, read the READ ME FIRST before doing anything

CALCULATE & UPDATE

RESET & RESTART

DOWNLOAD

Download

Once you're done, you can download the results in a .csv file ("," is the column separator)

The name of the file contains the label of the last D score you have computed

For instance, if $D3$ is the last algorithm you have computed, then the file name will be: ShinyAPPDscore3.csv

- 1 DScoreApp
- 2 **implicitMeasures**
- 3 Supplementary material

If you are an R user, `implicitMeasures` is made for you!

Advantages

- It's super easy to use
- Well documented and the maintainer is always available to help you
- You can compute multiple D scores all at once
- Computes the score also for the SC-IAT
- You can compute the D score and run further analysis all in the same place

Disadvantages

- You have to be familiar with R to efficiently use `implicitMeasures`

Install & Upload

Install package:

```
> install.packages("implicitMeasures")
```

Upload the package so you can use its functions:

```
> library(implicitMeasures)
```

and now you're good to go

Import dataset

Again, we use the data set obtained from Inquisit (file .dat):

```
> data = read.table("IATdata.dat", header = TRUE, sep = "\t")
> head(data)
```

	date	time	build	subject	blocknum	blockcode	trialnum	tr
1	121318	09:55	3.0.6.0	1	1	consenso	1	
2	121318	09:55	3.0.6.0	1	19	WhiteBlack	1	
3	121318	09:55	3.0.6.0	1	19	WhiteBlack	2	PlainW
4	121318	09:55	3.0.6.0	1	19	WhiteBlack	4	PlainBl
5	121318	09:55	3.0.6.0	1	19	WhiteBlack	6	PlainW
6	121318	09:55	3.0.6.0	1	19	WhiteBlack	8	PlainBl

	response
1	Disponibile a partecipare alla ricerca nei termini sopra indicati
2	57
3	18
4	23
5	18
6	23

blockcode

```
> table(data$blockcode)
```

badgood	consenso	demografica	PracticeWhite
5985	147	852	2
PracticeWhitegood	TestWhitebad	TestWhitegood	WhiteB
3003	5822	5863	3
WhiteBlack2nd			
3003			

We have a lot of stuff to get rid of. . . .

trialcode

```
> table(data$trialcode)
```

consenso	edu	età	occupazio	
147	142	142	142	
PlainBlackleft	PlainBlackright	Plaingoodright	PlainWhiteleft	Pla
3575	3561	7125	3560	
pol1	pol2	reminder	reminder1	
142	142	857	285	

Prepare data set

```
> data_clean = clean_iat(  
+   data, # data set name  
+   sbj_id = "subject", # column of sbj IDs  
+   block_id = "blockcode", # column of the block labels  
+   mapA_practice = "PracticeWhitegood",  
+   mapA_test = "TestWhitegood",  
+   mapB_practice = "PracticeWhitebad",  
+   mapB_test = "TestWhitebad",  
+   latency_id = "latency", # column with latency  
+   accuracy_id = "correct", # column with accuracy  
+   trial_id = "trialcode", # column with trial labels  
+   trial_eliminate = c("reminder", "reminder1"), # trials to get rid of  
+   demo_id = "blockcode", # column of the block labels  
+   trial_demo = "demografica" # label of the demographic trials  
+ )
```

```
> names(data_clean)
```

```
[1] "data_keep"      "data_eliminate" "demo"
```

- data_keep: the data set on which we compute the D scores (with class data.frame, iat_clean)
- data_eliminate: all the discarded trials
- demo: a data set containing all the demographic infos (we will see it later!)

```
> names(data_clean)
```

```
[1] "data_keep"      "data_eliminate" "demo"
```

- data_keep: the data set on which we compute the D scores (with class `data.frame`, `iat_clean`)
- data_eliminate: all the discarded trials
- demo: a data set containing all the demographic infos (we will see it later!)

If you export in `.csv` `data_keep` you can upload it to `DScoreApp`!

```
> write.table(data_clean[[1]], "cleanIAT.csv", sep = ",", row.names = FALSE)
```

Compute D score

```
> iat = data_clean[[1]]
```

Use the `compute_iat()` function:

```
> d3 = compute_iat(iat, Dscore = "d3")
```

```
> head(d3[, 1:5])
```

	participant	n_trial	nslow10000	nfast400	nfast300
1	1	120	0	0.01	0
2	2	120	0	0.03	0
3	3	120	0	0.14	0
4	4	120	0	0.07	0
5	5	120	0	0.00	0
6	6	120	0	0.00	0

```
> head(d3[, 6:10])
```

	accuracy.practice_MappingA	accuracy.practice_MappingB	accuracy.test_MappingA
1	1.00	1.00	
2	1.00	0.95	
3	0.95	0.75	
4	1.00	0.95	
5	0.95	1.00	
6	0.95	0.90	
	accuracy.test_MappingB	accuracy.MappingA	
1	0.950	0.9833333	
2	1.000	0.9833333	
3	0.900	0.9000000	
4	0.950	0.9666667	
5	0.925	0.9333333	
6	0.975	0.9166667	

```
> head(d3[, 11:15])
```

	accuracy.MappingB	RT_mean.MappingA	RT_mean.MappingB	mean_practice
1	0.9666667	597.6649	738.7075	
2	0.9833333	598.3324	649.2085	
3	0.8500000	575.2006	721.7637	
4	0.9500000	606.2957	645.4930	
5	0.9500000	849.8184	1011.9773	
6	0.9500000	914.6978	981.7482	

	mean_test_MappingA
1	585.7973
2	589.5736
3	585.2044
4	618.3685
5	695.3284
6	911.1501

```
> head(d3[, 16:19])
```

	mean_practice_MappingB	mean_test_MappingB	d_practice_d3	d_test_d3
1	851.5500	682.2862	1.00062841	0.6082550
2	840.5754	553.5250	0.62762704	-0.2318170
3	973.0089	596.1411	1.16298569	0.0550429
4	746.2660	595.1065	0.70572685	-0.1628654
5	1134.5500	950.6909	-0.04599615	0.6910860
6	1152.3929	896.4259	0.50668795	-0.0342550


```
> head(d3[, 20:21])
```

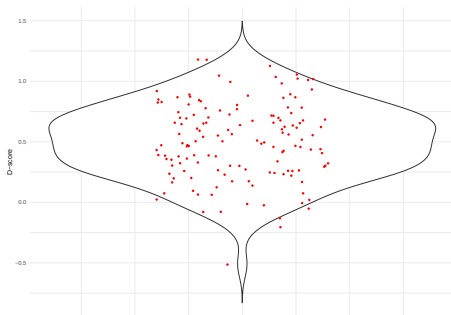
	dscore_d3	cond_ord
1	0.8044421	MappingB_First
2	0.1979047	MappingA_First
3	0.6090143	MappingB_First
4	0.2714307	MappingA_First
5	0.3225454	MappingA_First
6	0.2362163	MappingB_First

```
> head(d3[, 22:23])
```

	legendMappingA	legendMappingB
1	PracticeWhitegood_and_TestWhitegood	PracticeWhitebad_and_TestWhitegood
2	PracticeWhitegood_and_TestWhitegood	PracticeWhitebad_and_TestWhitegood
3	PracticeWhitegood_and_TestWhitegood	PracticeWhitebad_and_TestWhitegood
4	PracticeWhitegood_and_TestWhitegood	PracticeWhitebad_and_TestWhitegood
5	PracticeWhitegood_and_TestWhitegood	PracticeWhitebad_and_TestWhitegood
6	PracticeWhitegood_and_TestWhitegood	PracticeWhitebad_and_TestWhitegood

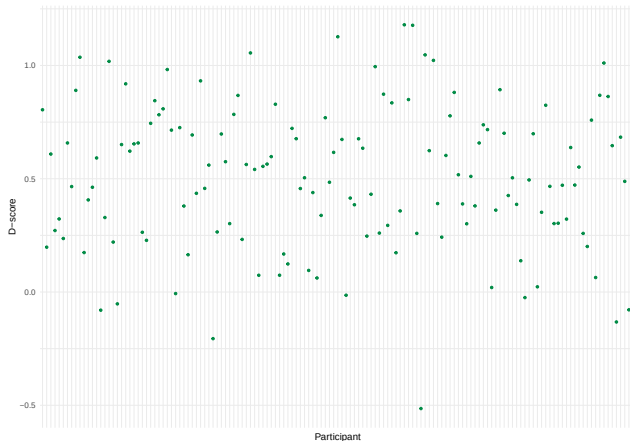
Some plots!

```
> d_density(d3, graph = "violin")
```



Other plots

```
> d_point(d3, x_values = FALSE)
```



Multiple D scores at once

```
> dscores = multi_dscores(iat, ds = "error-inflation")
```

Careful!

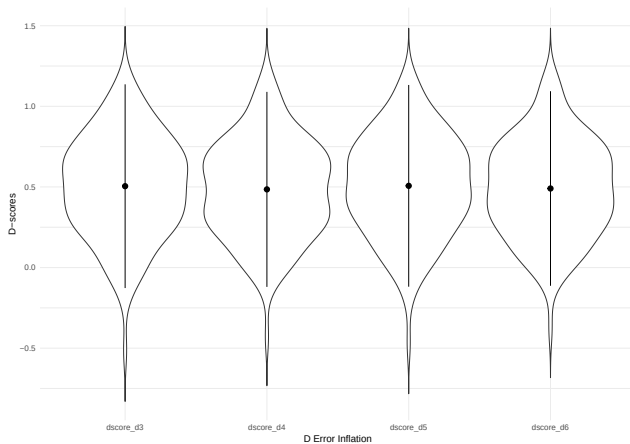
```
> names(dscores)
```

```
[1] "dscores" "graph"
```

```
> head(dscores[[1]])
```

	participant	dscore_d3	dscore_d4	dscore_d5	dscore_d6
1	1	0.8044421	0.7664824	0.7902660	0.75162963
2	2	0.1979047	0.1807094	0.1081013	0.09710327
3	3	0.6090143	0.5433544	0.6836311	0.65357652
4	4	0.2714307	0.2849834	0.4108129	0.43049456
5	5	0.3225454	0.3177612	0.3225454	0.31776123
6	6	0.2362163	0.2714998	0.2362163	0.27149977

```
> dscores[[2]]
```



Did we forget about demo...?

```
> demo_raw = data_clean[[3]]
> str(demo_raw)
```

```
'data.frame':   852 obs. of  12 variables:
 $ date       : int   121318 121318 121318 121318 121318 121318 121318 121318 121318 121318 121318 121318
 $ time       : chr    "09:55" "09:55" "09:55" "09:55" ...
 $ build      : chr    "3.0.6.0" "3.0.6.0" "3.0.6.0" "3.0.6.0" ...
 $ participant : int    1 1 1 1 1 1 2 2 2 2 ...
 $ blocknum   : int    57 57 57 57 57 57 57 57 57 57 ...
 $ blockcode  : chr    "demografica" "demografica" "demografica" "demografica" ...
 $ trialnum   : int    1 1 1 1 4 4 1 1 1 1 ...
 $ trialcode  : chr    "sesso" "età" "occupazio" "edu" ...
 $ response   : chr    "Maschio" "21" "stud" "sup" ...
 $ correct    : int    1 1 1 1 1 1 1 1 1 1 ...
 $ latency    : int   19185 19185 19185 19185 28866 28866 24586 24586 ...
 $ stimulusitem1: chr    "0" "0" "0" "0" ...
....
```


Let's start by keeping only the column of interest:

```
> demo_raw = demo_raw[, c("participant", "trialcode", "response")]  
> str(demo_raw)
```

```
'data.frame':  852 obs. of  3 variables:  
 $ participant: int  1 1 1 1 1 1 2 2 2 2 ...  
 $ trialcode  : chr  "sesso" "età" "occupazio" "edu" ...  
 $ response   : chr  "Maschio" "21" "stud" "sup" ...
```

Reshape the data set

```
> demo <- reshape(demo_raw, timevar = "trialcode", idvar = "pa
> str(demo)
```

```
'data.frame':  142 obs. of  7 variables:
 $ participant      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ response.sesso   : chr  "Maschio" "Maschio" "Femmina" "Fem
 $ response.età     : chr  "21" "31" "21" "21" ...
 $ response.occupazio: chr  "stud" "stud" "stud" "stud" ...
 $ response.edu     : chr  "sup" "magistrale" "sup" "sup" ...
 $ response.pol1    : chr  "3" "3" "3" "2" ...
 $ response.pol2    : chr  "2" "5" "2" "3" ...
....
```

Polish the data set

```
> colnames(demo) <- gsub("response.", "", colnames(demo))
> demo[, c(3, 6:7)] <- apply(demo[, c(3, 6:7)], 2, as.integer)
> str(demo)

'data.frame':   142 obs. of  7 variables:
 $ participant: int  1 2 3 4 5 6 7 8 9 10 ...
 $ sesso      : chr  "Maschio" "Maschio" "Femmina" "Femmina" ...
 $ età       : int  21 31 21 21 21 20 20 19 20 20 ...
 $ occupazio  : chr  "stud" "stud" "stud" "stud" ...
 $ edu        : chr  "sup" "magistrale" "sup" "sup" ...
 $ pol1       : int  3 3 3 2 4 3 2 3 3 3 ...
 $ pol2       : int  2 5 2 3 4 2 3 2 1 2 ...
....
```

Merge demo with d3

```
> d3complete = merge(d3, demo, by = "participant")
> str(d3complete[17:29])
```

```
'data.frame':  142 obs. of  13 variables:
 $ mean_test_MappingB: num  682 554 596 595 951 ...
 $ d_practice_d3      : num  1.001 0.628 1.163 0.706 -0.046 ...
 $ d_test_d3         : num  0.608 -0.232 0.055 -0.163 0.691 ...
 $ dscore_d3         : num  0.804 0.198 0.609 0.271 0.323 ...
 $ cond_ord          : chr   "MappingB_First" "MappingA_First" "MappingB_Second" ...
 $ legendMappingA     : chr   "PracticeWhitegood_and_TestWhitegood" "PracticeWhitebad_and_TestWhitebad" ...
 $ legendMappingB     : chr   "PracticeWhitebad_and_TestWhitebad" "PracticeWhitegood_and_TestWhitegood" ...
 $ sesso              : chr   "Maschio" "Maschio" "Femmina" "Femmina" ...
 $ età                : int   21 31 21 21 21 20 20 19 20 20 ...
 $ occupazio          : chr   "stud" "stud" "stud" "stud" ...
 $ edu                : chr   "sup" "magistrale" "sup" "sup" ...
 $ pol1               : int   3 3 3 2 4 3 2 3 3 3 ...
 $ pol2               : int   2 5 2 3 4 2 3 2 1 2 ...
```

Compute correlation

```
> correlations <- data.frame(cor(d3complete[, c("dscore_d3", "pol1")
> correlations <- round(correlations, 2)
> correlations[upper.tri(correlations, diag = TRUE)] <- ""
```

Results are in Table 1:

Table 1: Race IAT correlations

	dscore_d3	pol1	pol2
dscore_d3			
pol1	0		
pol2	0.12	-0.34	

- 1 DScoreApp
- 2 implicitMeasures
- 3 **Supplementary material**

DScoreApp:

Epifania, O. M., Anselmi, P., & Robusto, E. (2020). Dscoreapp: A shiny web application for the computation of the implicit association test D score. Frontiers in Psychology, 10, 2938. doi: 10.3389/fpsyg.2019.02938

implicitMeasures:

Epifania, O. M., Anselmi, P., & Robusto, E. (2020). Implicit measures with reproducible results: The implicitmeasures package. Journal of Open Source Software, 5(52), 2394. doi: 10.21105/joss.02394